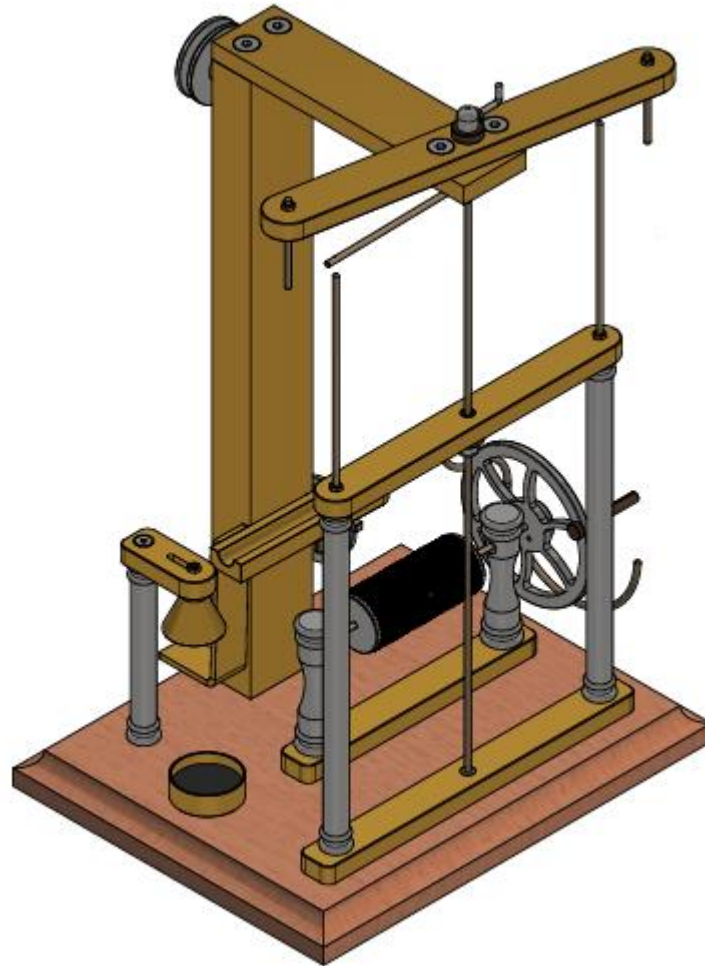


Automatic Tea Steeper Contraption

Design Overview



Design by
Tom Dunbar



This is a design for an automatic tea steeping contraption that is gravity powered and steeps a tea bag for approximately 5 minutes. After the steeping is complete, a bell is rung one time and the tea bag is lifted from the water. The contraption is based around a flying pendulum escapement so it contains no gears, despite it essentially being a clock.

The contraption was inspired by this delightfully over-engineered tea temperature monitoring mechanism:

<https://www.youtube.com/watch?v=oJzy1vk2zyc&list=PLxYRkul9JZoOfj6--7C8BmSLWSMhE7KK8&index=25&pp=gAQBiAQB>

From this video, I liked the notification mechanism so much, I incorporated it into my design. I also like the aesthetic and functionality of the Kontax Stirling engine (<https://stirlingengine.co.uk/products/KS90R-Ross-yoke-Aluminium-Stirling-Engine-p593670073>), so I used some parts of it in the design. The goal is to create something that would look at home in a Victorian inventor's parlor. As such, I did not want to use any electronics and wanted to use a 'natural' energy source e.g. gravity or the heat from the hot water.



Building a Bimetallic Tea Monitoring Mechanism



Main Mechanism Overview

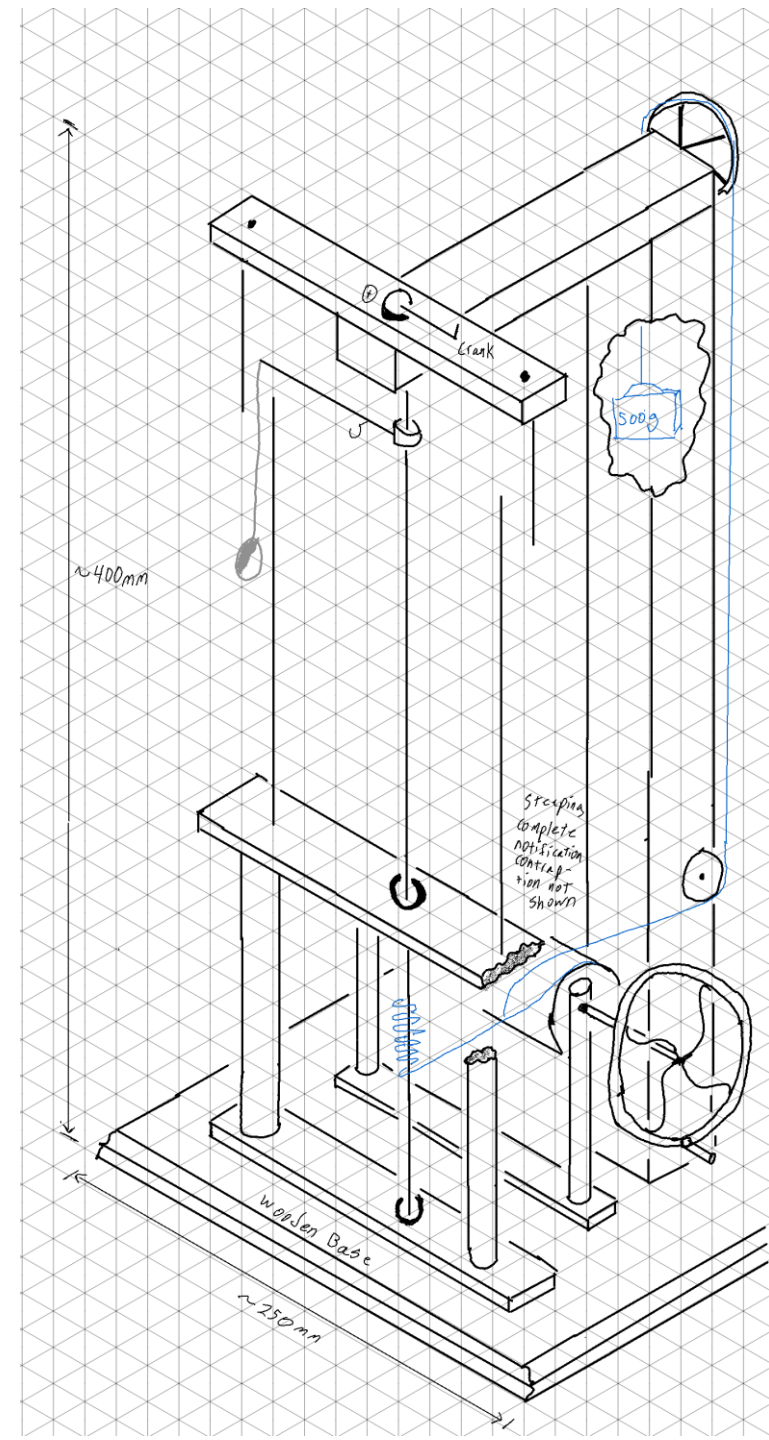
The mechanism works by gravity pulling the weight down, pulling on the string (blue).

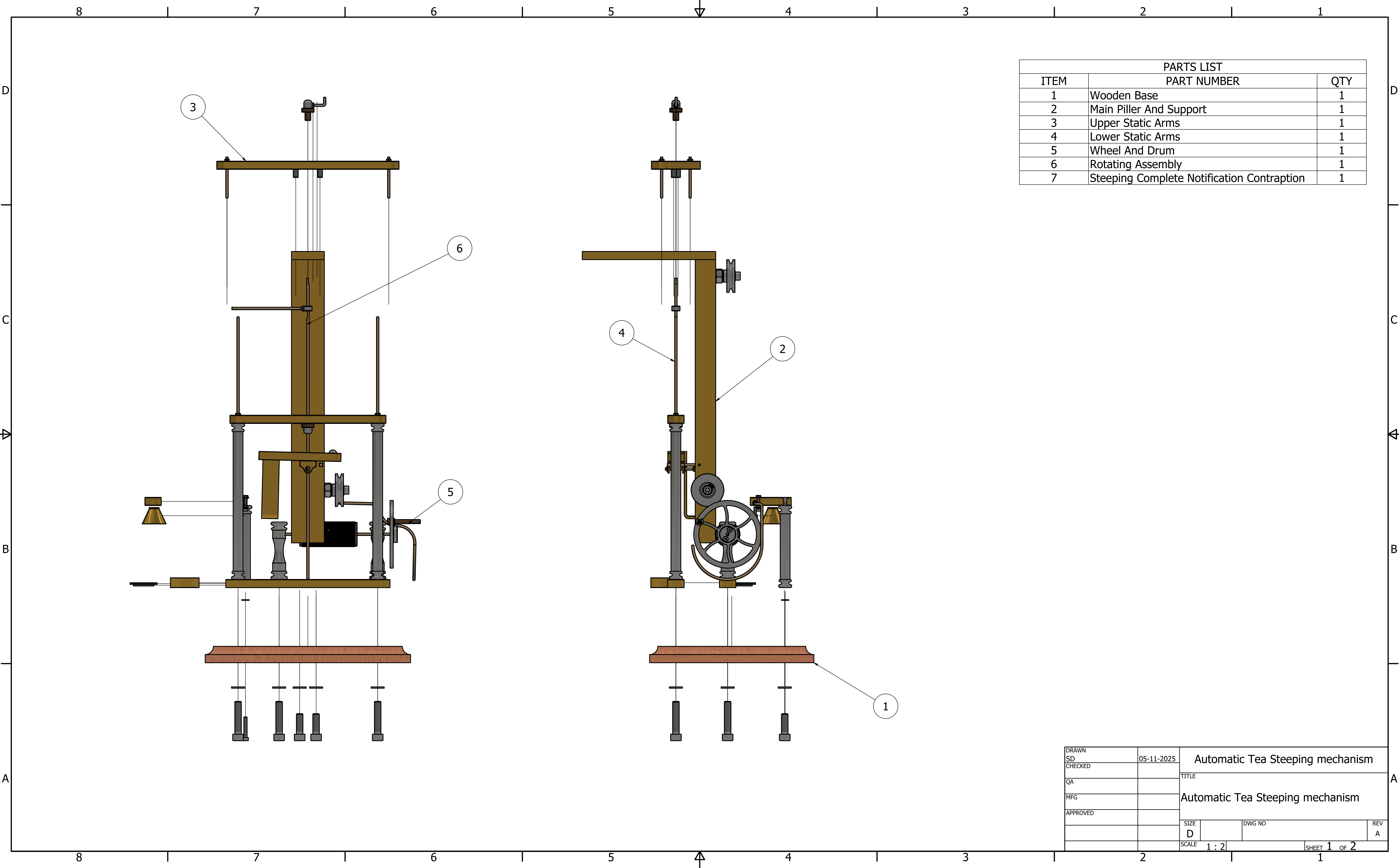
The string is wrapped around a main vertical shaft and a drum. The drum is on a horizontal shaft with a wheel. As the string is played out, it causes the main vertical shaft to rotate as well as the drum/wheel to rotate.

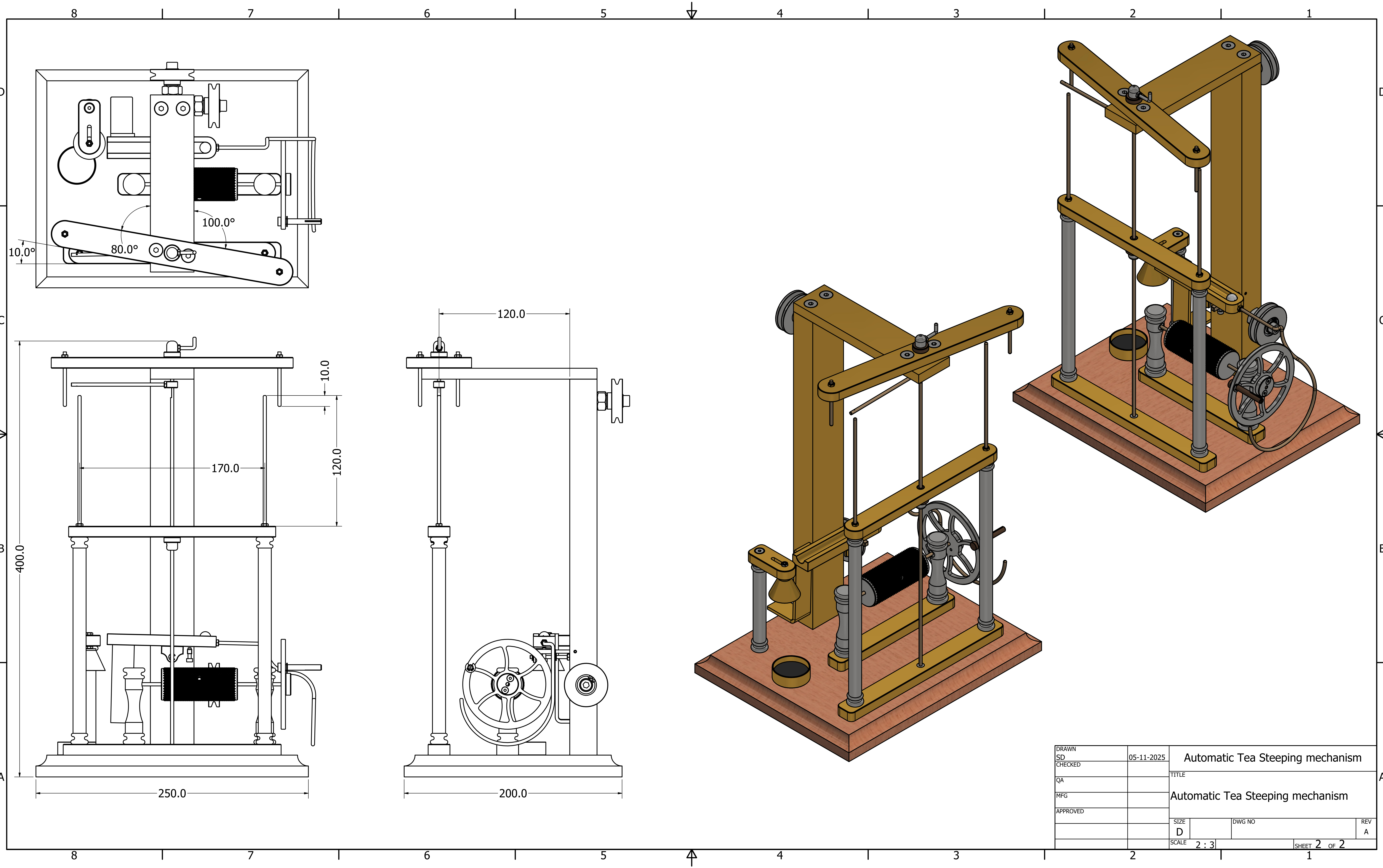
The main shaft has a flying pendulum escapement to slow the weight's decent.

A tea bag's string is attached to a peg on the wheel and a mug of hot water is positioned below the wheel. As the wheel rotates, it slowly steeps the tea bag up and down.

As discussed later, after 5 minutes there is a steeping complete notification that rings a bell one time to alert the tea drinker that the tea is ready, similar to the notification in the inspiration video. The mechanism also lifts the tea bag out of the water.





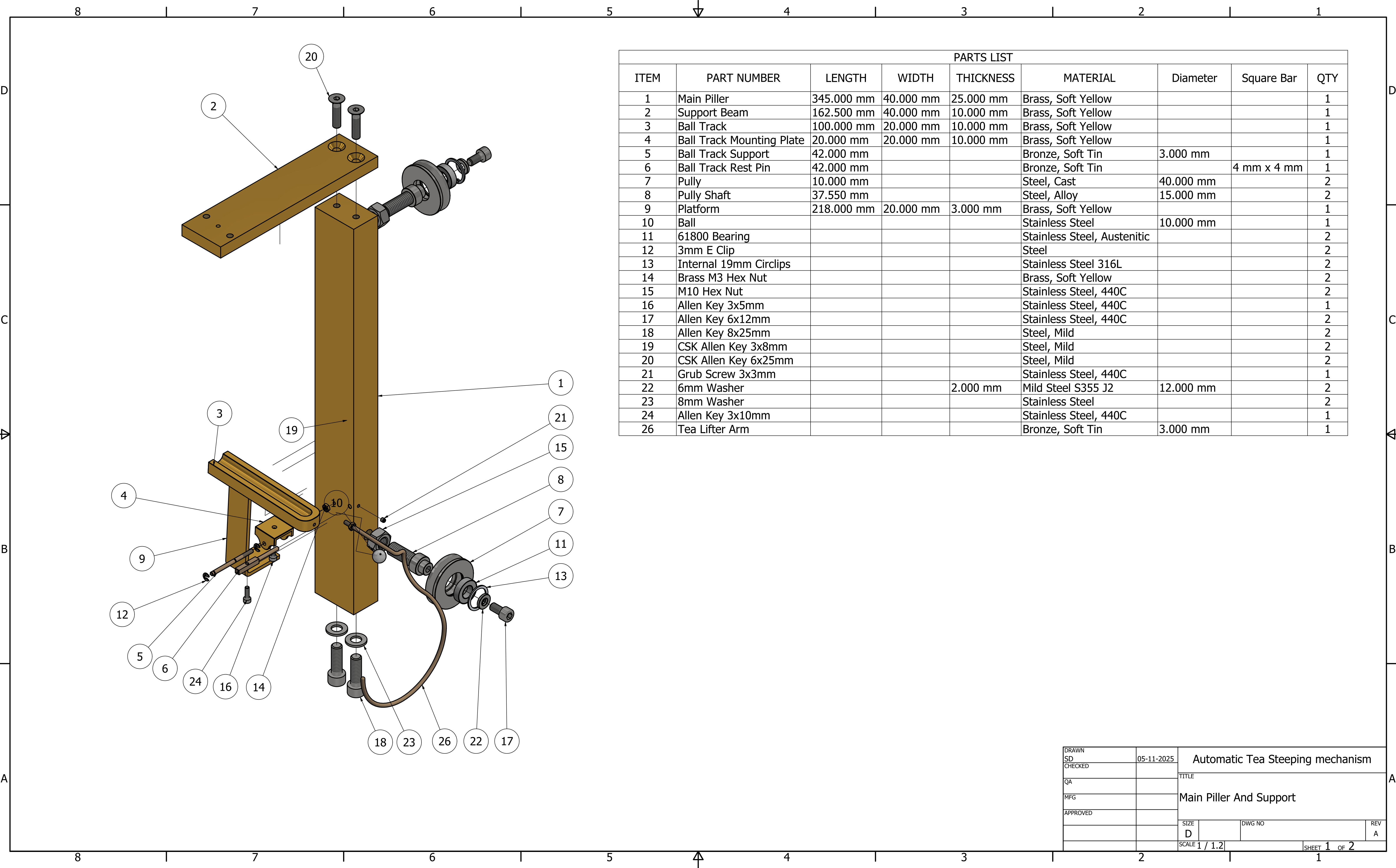


Flying Pendulum Escapement

- The link below is a good introduction to a flying pendulum and discusses some of the critical dimensions:
 - <https://www.youtube.com/watch?v=z8V6RMJbHJc>
- The link below provides some good technical drawings of the escapement and some of the dimensions are visible in the PDF that you can view online. I used some of these dimensions when designing the device.
 - <https://www.woodenclocks.co.uk/clock-17/>
- The following video is a bit long, but provided the original a-ha moment for how to actually accomplish the auto-steeping mechanism. The main mechanism is essentially a re-arrangement of his contraption:
 - <https://www.youtube.com/watch?v=uuelh5Yt9-Q>

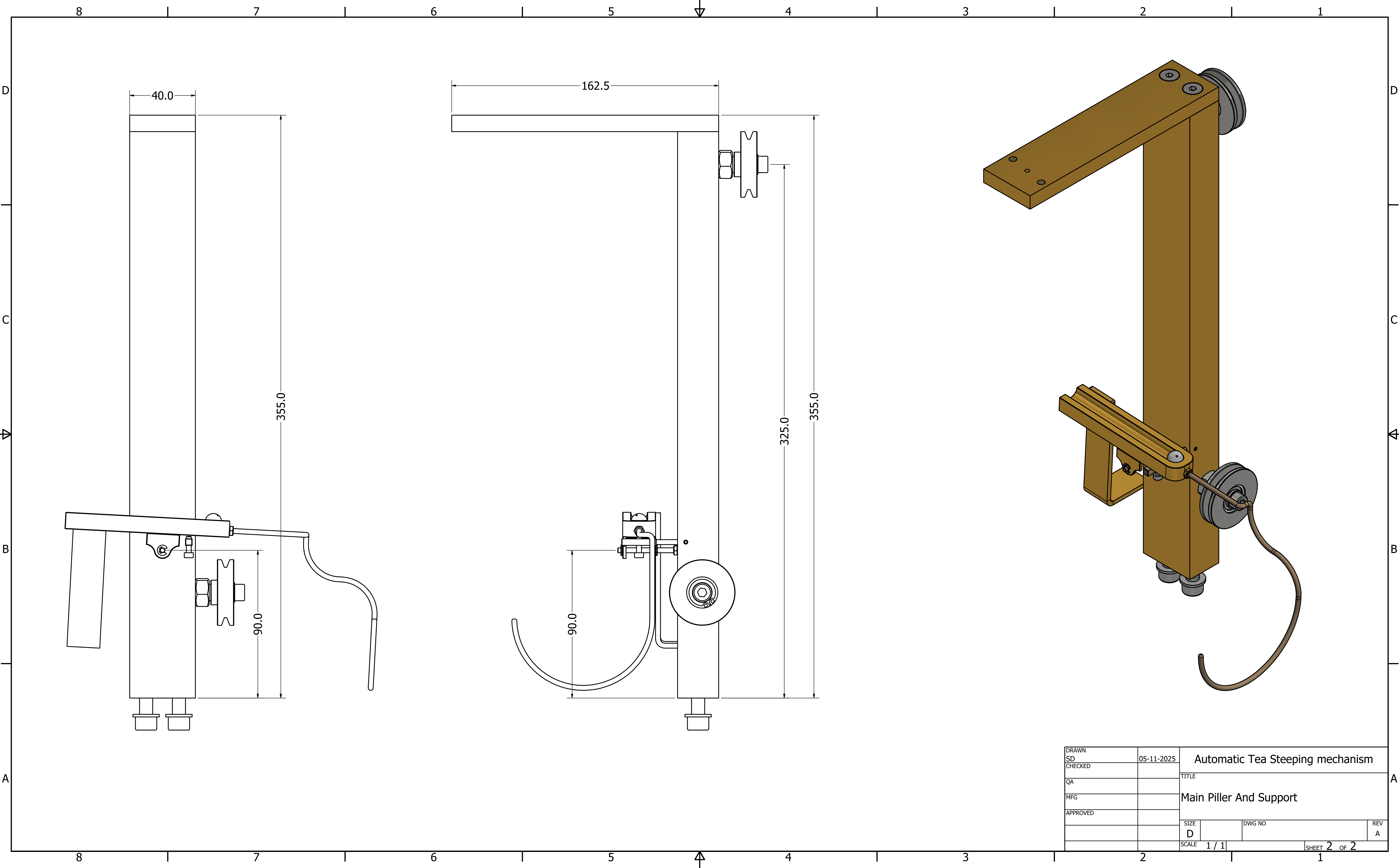
Sub Assemblies

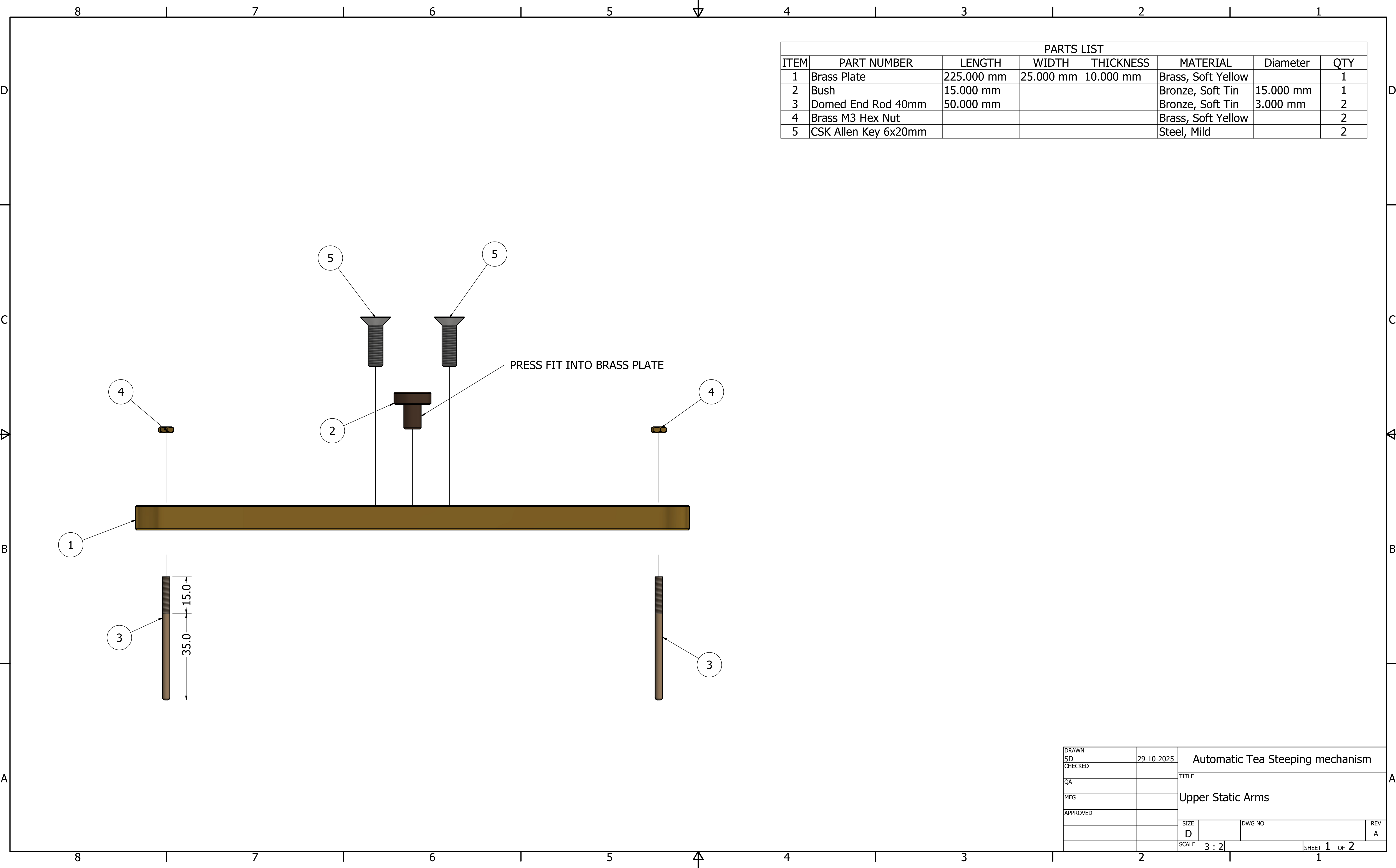
- Main pillar and Support
- Upper static arms
- Lower static arms
- Rotating assembly
- Wheel & Drum
- Steeping complete notification contraption
- Wooden Base
- String and Weight

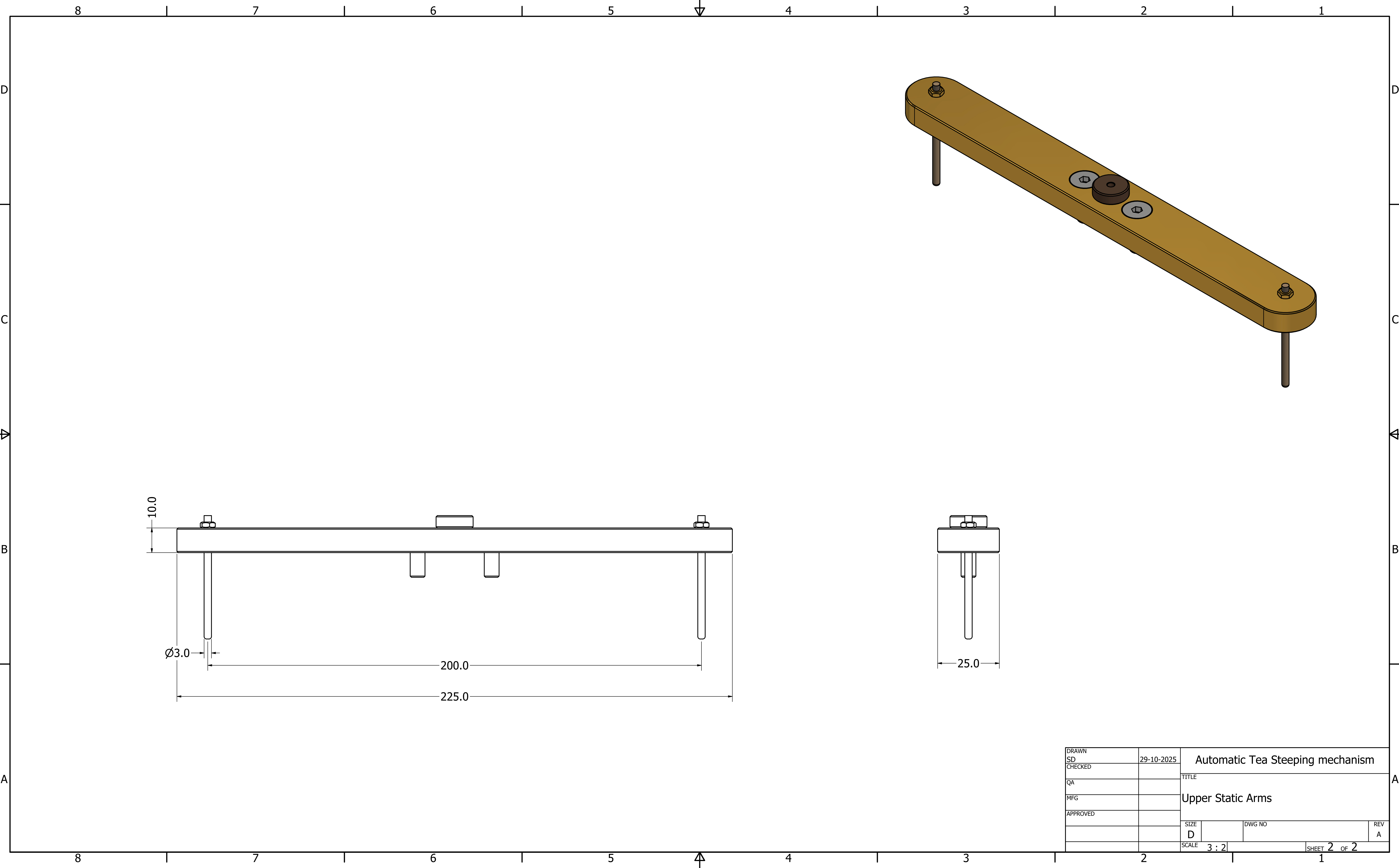


PARTS LIST								
ITEM	PART NUMBER	LENGTH	WIDTH	THICKNESS	MATERIAL	Diameter	Square Bar	QTY
1	Main Piller	345.000 mm	40.000 mm	25.000 mm	Brass, Soft Yellow			1
2	Support Beam	162.500 mm	40.000 mm	10.000 mm	Brass, Soft Yellow			1
3	Ball Track	100.000 mm	20.000 mm	10.000 mm	Brass, Soft Yellow			1
4	Ball Track Mounting Plate	20.000 mm	20.000 mm	10.000 mm	Brass, Soft Yellow			1
5	Bail Track Support	42.000 mm			Bronze, Soft Tin	3.000 mm		1
6	Ball Track Rest Pin	42.000 mm			Bronze, Soft Tin		4 mm x 4 mm	1
7	Pully	10.000 mm			Steel, Cast	40.000 mm		2
8	Pully Shaft	37.550 mm			Steel, Alloy	15.000 mm		2
9	Platform	218.000 mm	20.000 mm	3.000 mm	Brass, Soft Yellow			1
10	Ball				Stainless Steel	10.000 mm		1
11	61800 Bearing				Stainless Steel, Austenitic			2
12	3mm E Clip				Steel			2
13	Internal 19mm Circlips				Stainless Steel 316L			2
14	Brass M3 Hex Nut				Brass, Soft Yellow			2
15	M10 Hex Nut				Stainless Steel, 440C			2
16	Allen Key 3x5mm				Stainless Steel, 440C			1
17	Allen Key 6x12mm				Stainless Steel, 440C			2
18	Allen Key 8x25mm				Steel, Mild			2
19	CSK Allen Key 3x8mm				Steel, Mild			2
20	CSK Allen Key 6x25mm				Steel, Mild			2
21	Grub Screw 3x3mm				Stainless Steel, 440C			1
22	6mm Washer			2.000 mm	Mild Steel S355 J2	12.000 mm		2
23	8mm Washer				Stainless Steel			2
24	Allen Key 3x10mm				Stainless Steel, 440C			1
26	Tea Lifter Arm				Bronze, Soft Tin	3.000 mm		1

DRAWN	SD	05-11-2025	Automatic Tea Steeping mechanism		
CHECKED			TITLE		
QA			Main Piller And Support		
MFG					
APPROVED			SIZE	DWG NO	REV
			D		A
		SCALE 1 / 1.2	SHEET 1 OF 2		

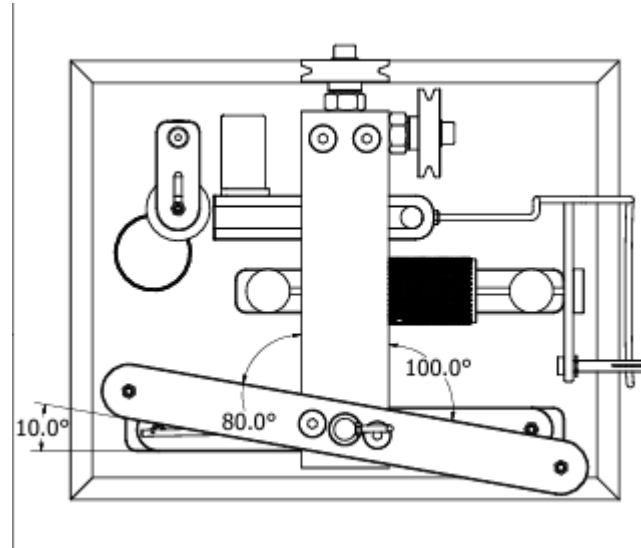


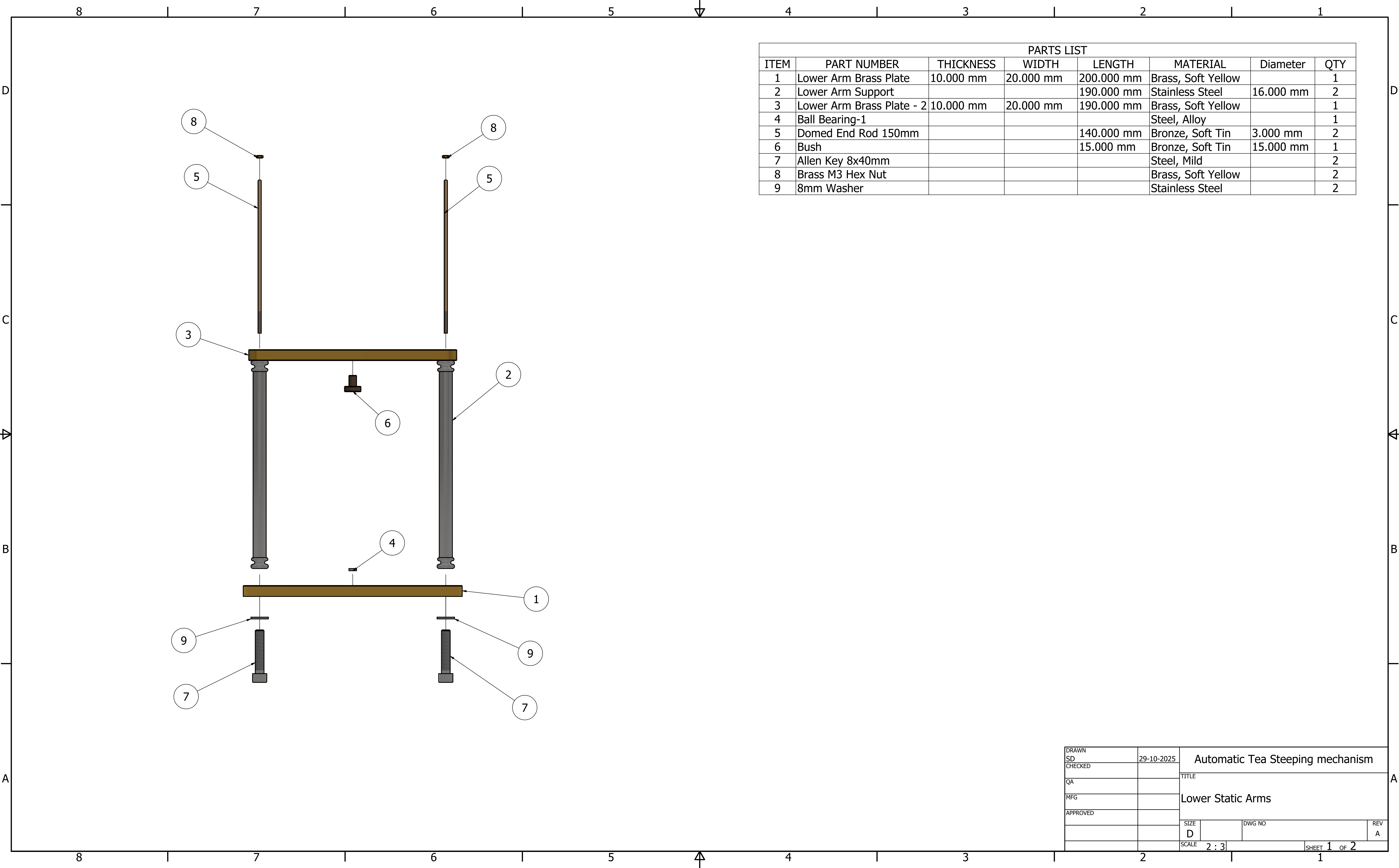




A slight offset (~ 10 degrees) between the upper and lower arms allows the string to disengage cleanly and helps it engage on the upper arm first when it swings around.

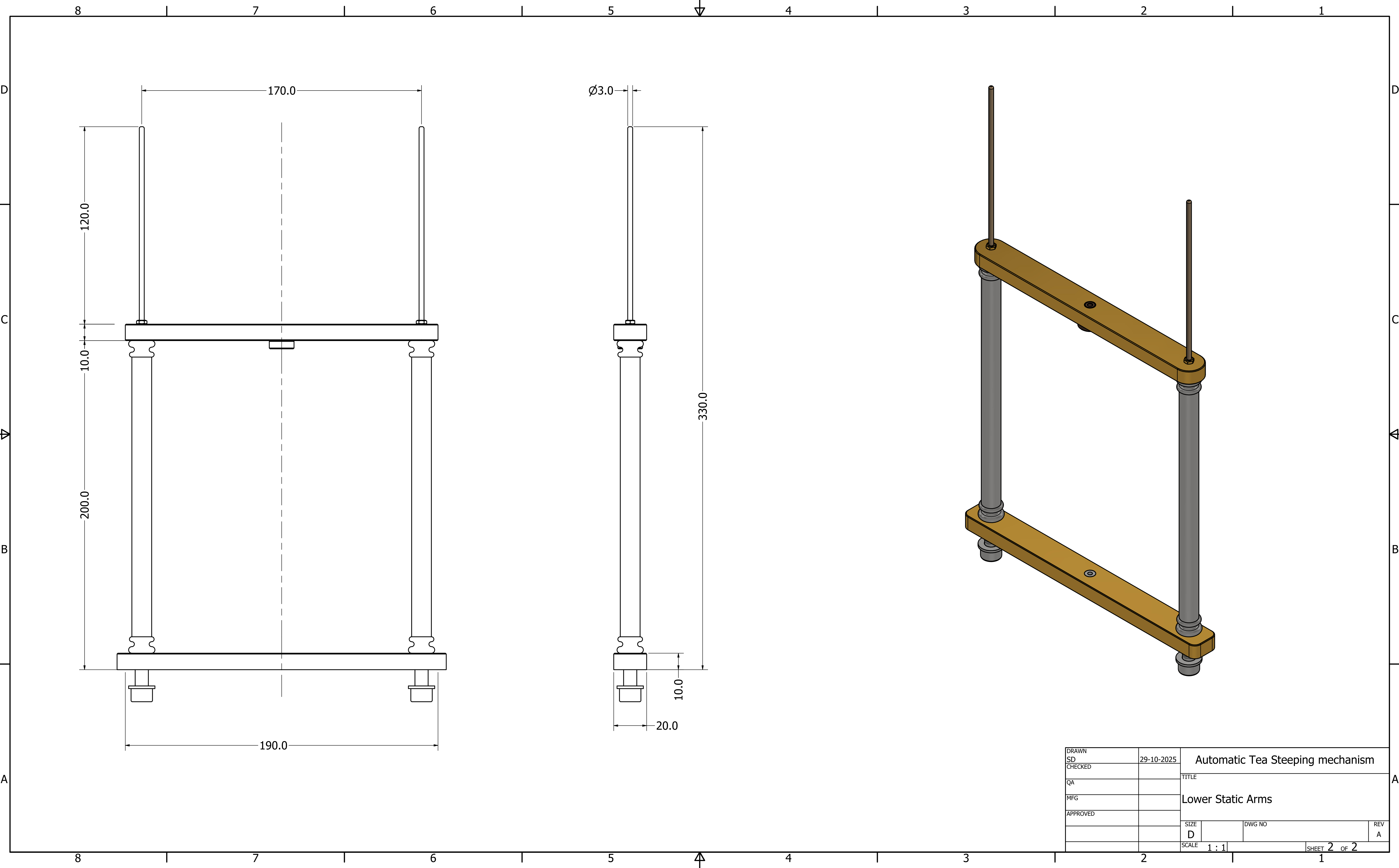
In this view, the pendulum is rotating counter-clockwise.

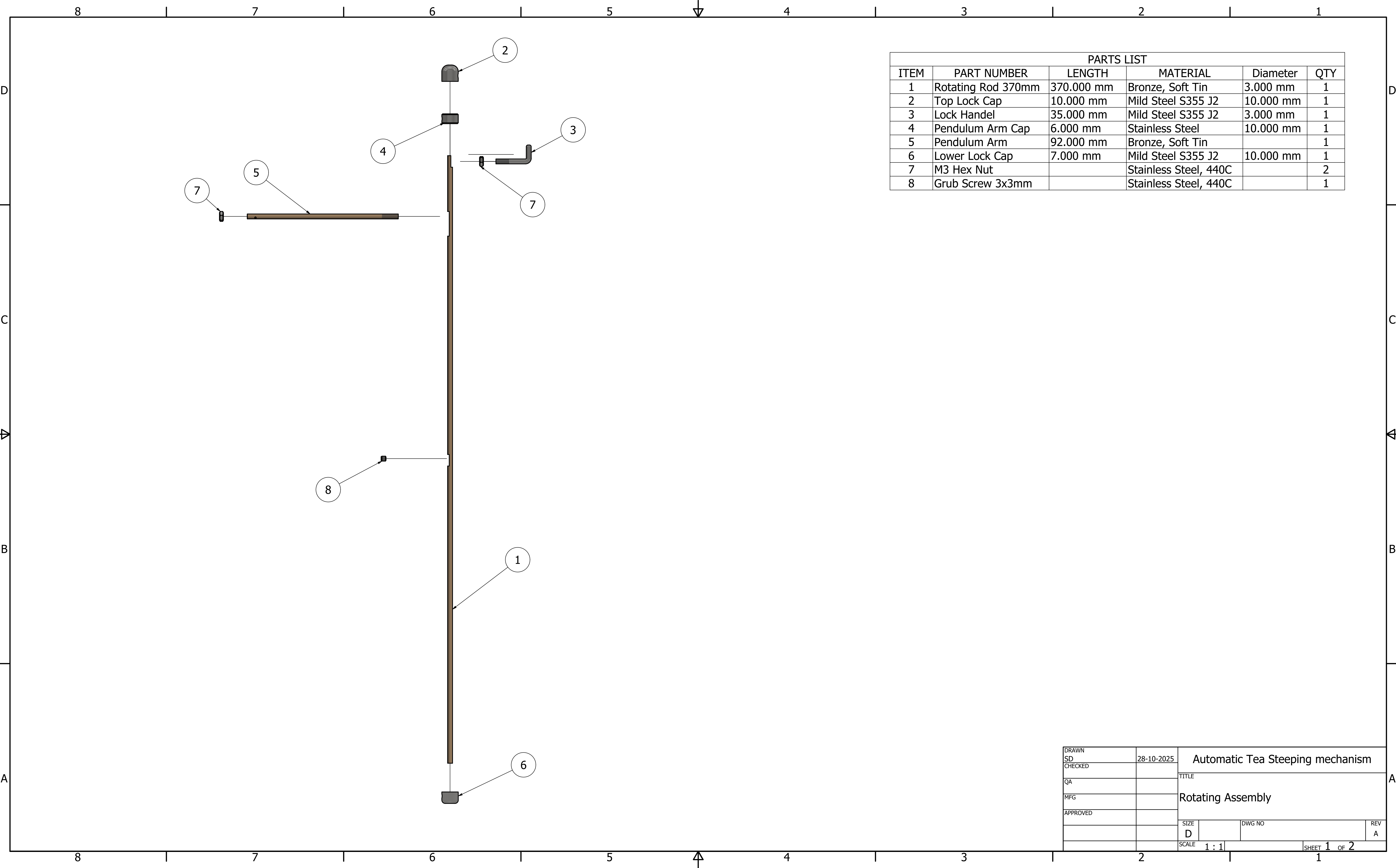


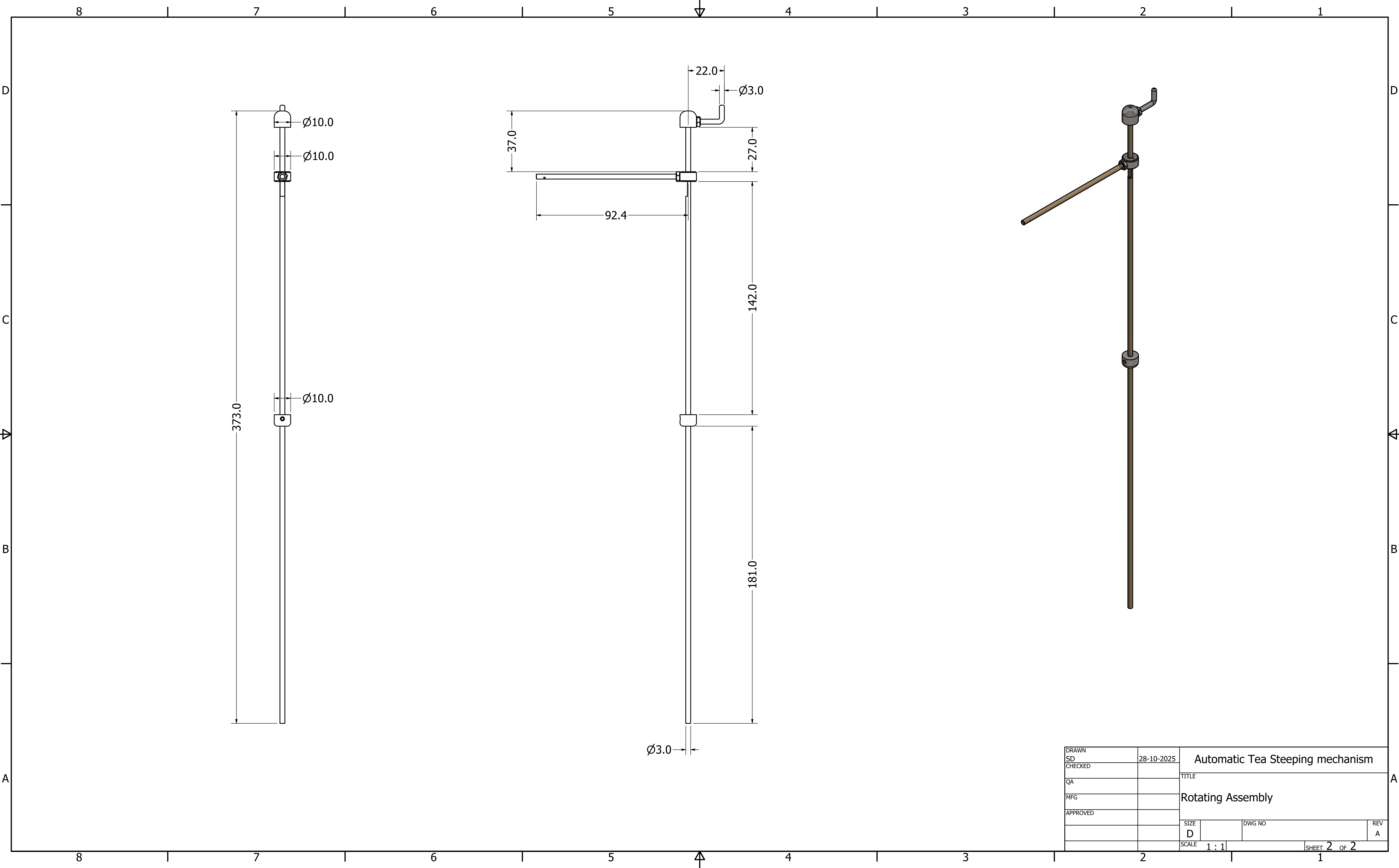


PARTS LIST							
ITEM	PART NUMBER	THICKNESS	WIDTH	LENGTH	MATERIAL	Diameter	QTY
1	Lower Arm Brass Plate	10.000 mm	20.000 mm	200.000 mm	Brass, Soft Yellow		1
2	Lower Arm Support			190.000 mm	Stainless Steel	16.000 mm	2
3	Lower Arm Brass Plate - 2	10.000 mm	20.000 mm	190.000 mm	Brass, Soft Yellow		1
4	Ball Bearing-1				Steel, Alloy		1
5	Domed End Rod 150mm			140.000 mm	Bronze, Soft Tin	3.000 mm	2
6	Bush			15.000 mm	Bronze, Soft Tin	15.000 mm	1
7	Allen Key 8x40mm				Steel, Mild		2
8	Brass M3 Hex Nut				Brass, Soft Yellow		2
9	8mm Washer				Stainless Steel		2

DRAWN	SD	29-10-2025	Automatic Tea Steeping mechanism		
CHECKED			TITLE		
QA			Lower Static Arms		
MFG					
APPROVED					
			SIZE	DWG NO	REV
			D		A
			SCALE	2 : 3	SHEET 1 OF 2







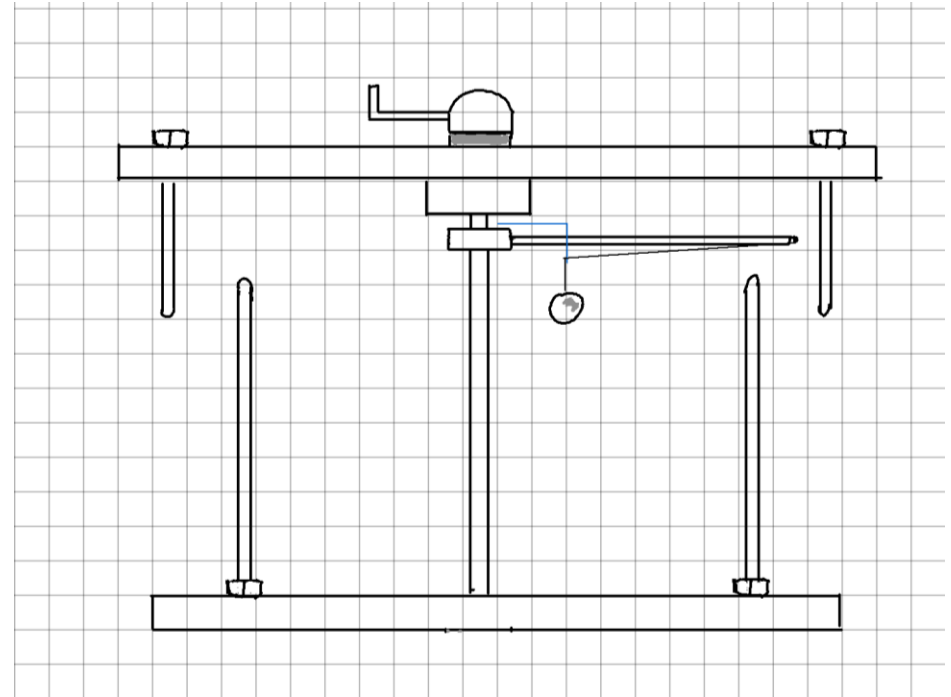
Resetting the device

To reset the device, the flying pendulum bob is placed onto the wire hook that is on the pendulum arm. This allows the user to spin the crank without the pendulum bob wrapping around the static arms. This will wind up the string and prepare the weight for its next decent.

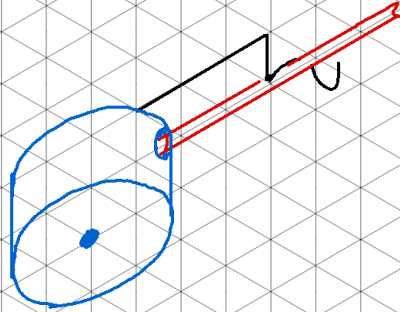
The pendulum bob is then looped over the upper static arm support beam to prevent the pendulum arm from moving (as seen in this video: <https://www.youtube.com/watch?v=z8V6RMJbHJc>)

When the pendulum is prevented from moving, the user can attach a tea bag's string to the peg on the wheel and prepare a mug of hot water.

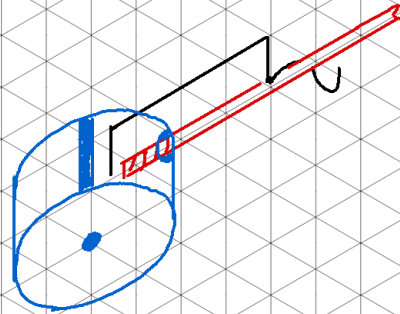
Un-looping the pendulum bob and letting go is all that is needed to start the main mechanism.

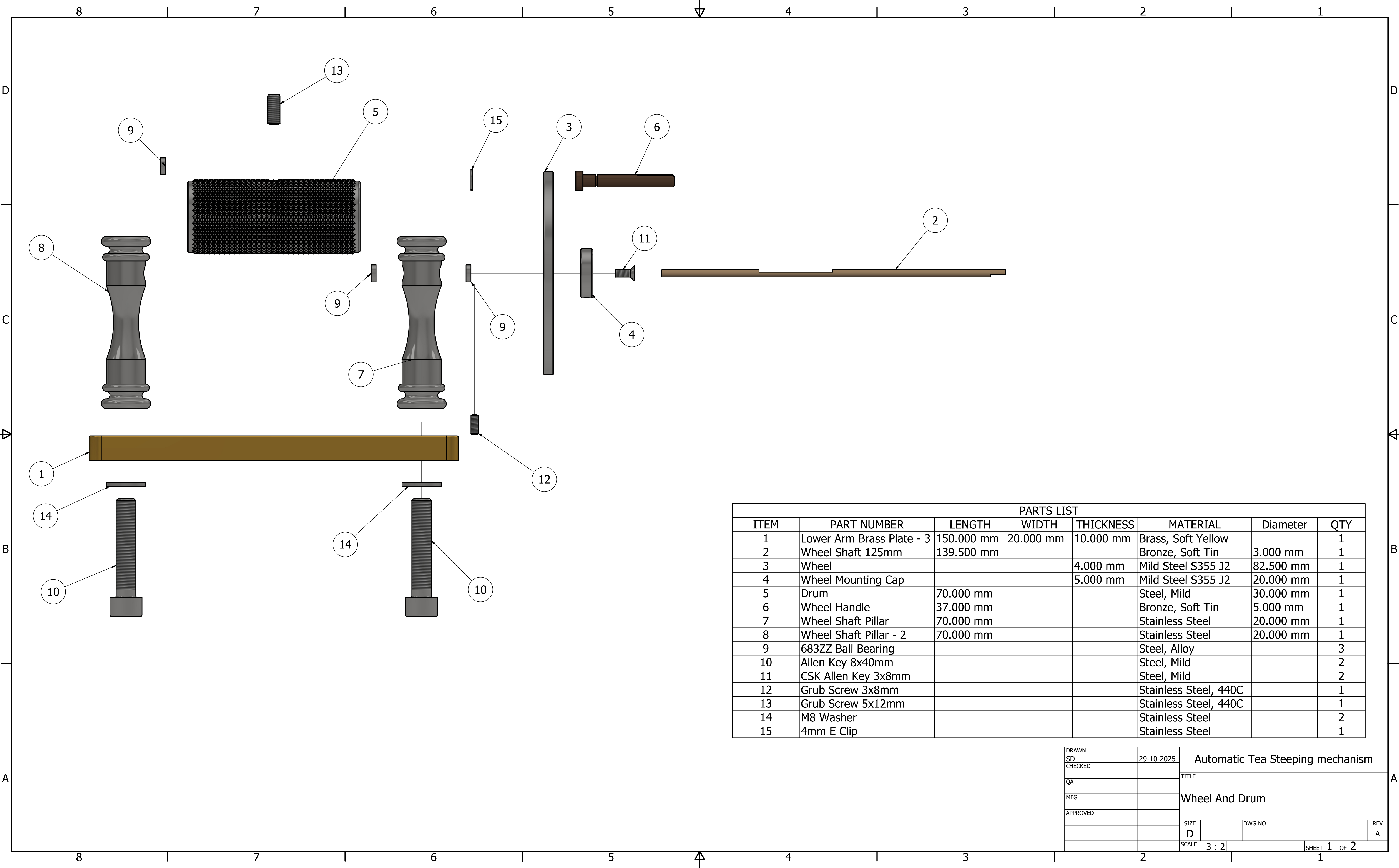


Wire hook
Isometric looking up



Isometric looking up and
x-ray vision



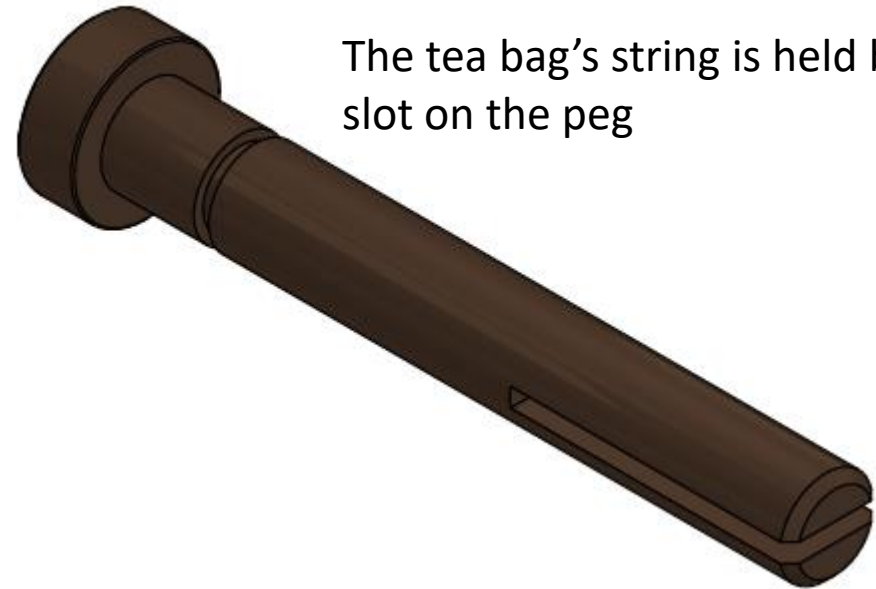


PARTS LIST							
ITEM	PART NUMBER	LENGTH	WIDTH	THICKNESS	MATERIAL	Diameter	QTY
1	Lower Arm Brass Plate - 3	150.000 mm	20.000 mm	10.000 mm	Brass, Soft Yellow		1
2	Wheel Shaft 125mm	139.500 mm			Bronze, Soft Tin	3.000 mm	1
3	Wheel			4.000 mm	Mild Steel S355 J2	82.500 mm	1
4	Wheel Mounting Cap			5.000 mm	Mild Steel S355 J2	20.000 mm	1
5	Drum	70.000 mm			Steel, Mild	30.000 mm	1
6	Wheel Handle	37.000 mm			Bronze, Soft Tin	5.000 mm	1
7	Wheel Shaft Pillar	70.000 mm			Stainless Steel	20.000 mm	1
8	Wheel Shaft Pillar - 2	70.000 mm			Stainless Steel	20.000 mm	1
9	683ZZ Ball Bearing				Steel, Alloy		3
10	Allen Key 8x40mm				Steel, Mild		2
11	CSK Allen Key 3x8mm				Steel, Mild		2
12	Grub Screw 3x8mm				Stainless Steel, 440C		1
13	Grub Screw 5x12mm				Stainless Steel, 440C		1
14	M8 Washer				Stainless Steel		2
15	4mm E Clip				Stainless Steel		1

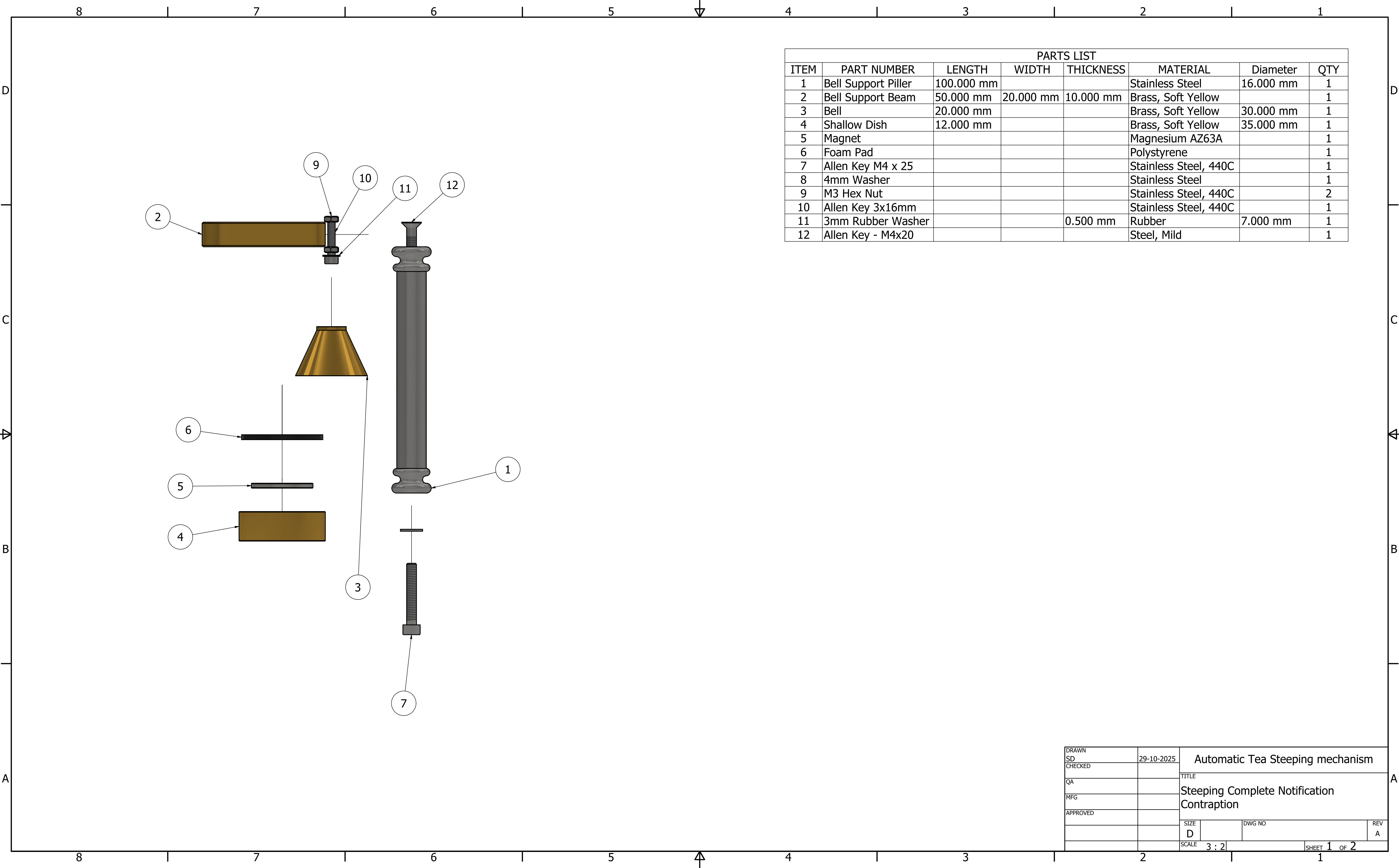
DRAWN	SD	29-10-2025	Automatic Tea Steeping mechanism			
CHECKED			TITLE			
QA			Wheel And Drum			
MFG						
APPROVED						
			SIZE		DWG NO	REV
			D			A
			SCALE	3 : 2		SHEET 1 OF 2

Wheel and drum

Wheel is inspired by the Kontax Stirling Engine wheel

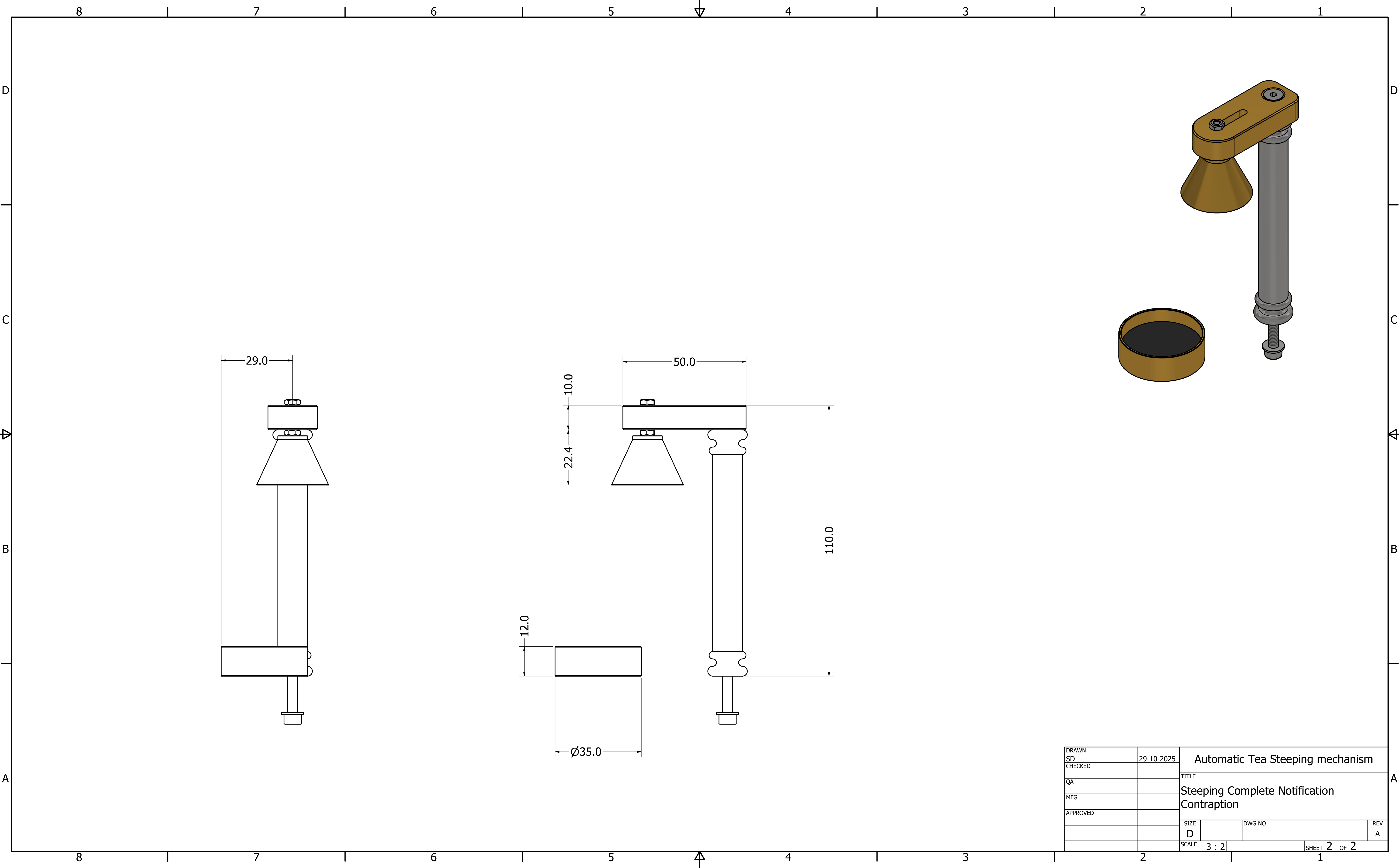


The tea bag's string is held by the slot on the peg



PARTS LIST							
ITEM	PART NUMBER	LENGTH	WIDTH	THICKNESS	MATERIAL	Diameter	QTY
1	Bell Support Piller	100.000 mm			Stainless Steel	16.000 mm	1
2	Bell Support Beam	50.000 mm	20.000 mm	10.000 mm	Brass, Soft Yellow		1
3	Bell	20.000 mm			Brass, Soft Yellow	30.000 mm	1
4	Shallow Dish	12.000 mm			Brass, Soft Yellow	35.000 mm	1
5	Magnet				Magnesium AZ63A		1
6	Foam Pad				Polystyrene		1
7	Allen Key M4 x 25				Stainless Steel, 440C		1
8	4mm Washer				Stainless Steel		1
9	M3 Hex Nut				Stainless Steel, 440C		2
10	Allen Key 3x16mm				Stainless Steel, 440C		1
11	3mm Rubber Washer			0.500 mm	Rubber	7.000 mm	1
12	Allen Key - M4x20				Steel, Mild		1

DRAWN	SD	29-10-2025	Automatic Tea Steeping mechanism			
CHECKED			TITLE			
QA			Steeping Complete Notification Contraption			
MFG						
APPROVED						
			SIZE		DWG NO	REV
			D			A
			SCALE	3 : 2		SHEET 1 OF 2

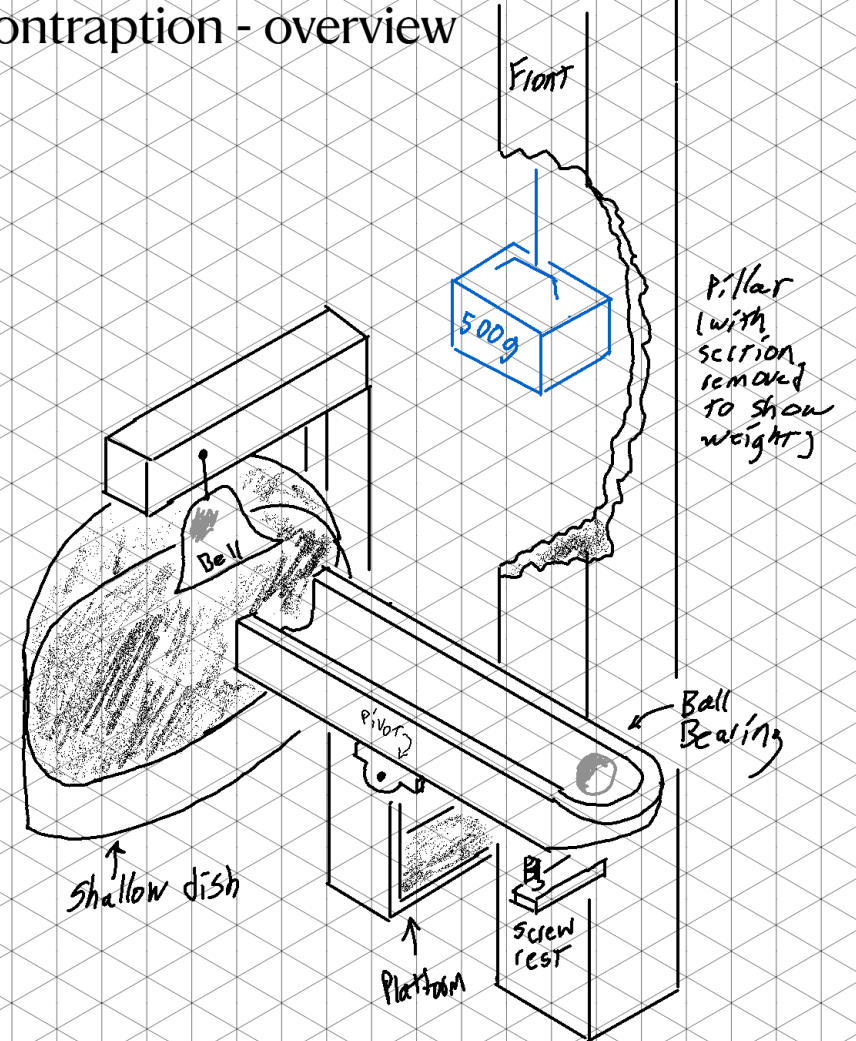


DRAWN	SD	29-10-2025	Automatic Tea Steeping mechanism		
CHECKED			TITLE		
QA			Steeping Complete Notification		
MFG			Contraction		
APPROVED			SIZE	DWG NO	REV
			D		A
			SCALE	3 : 2	SHEET 2 OF 2

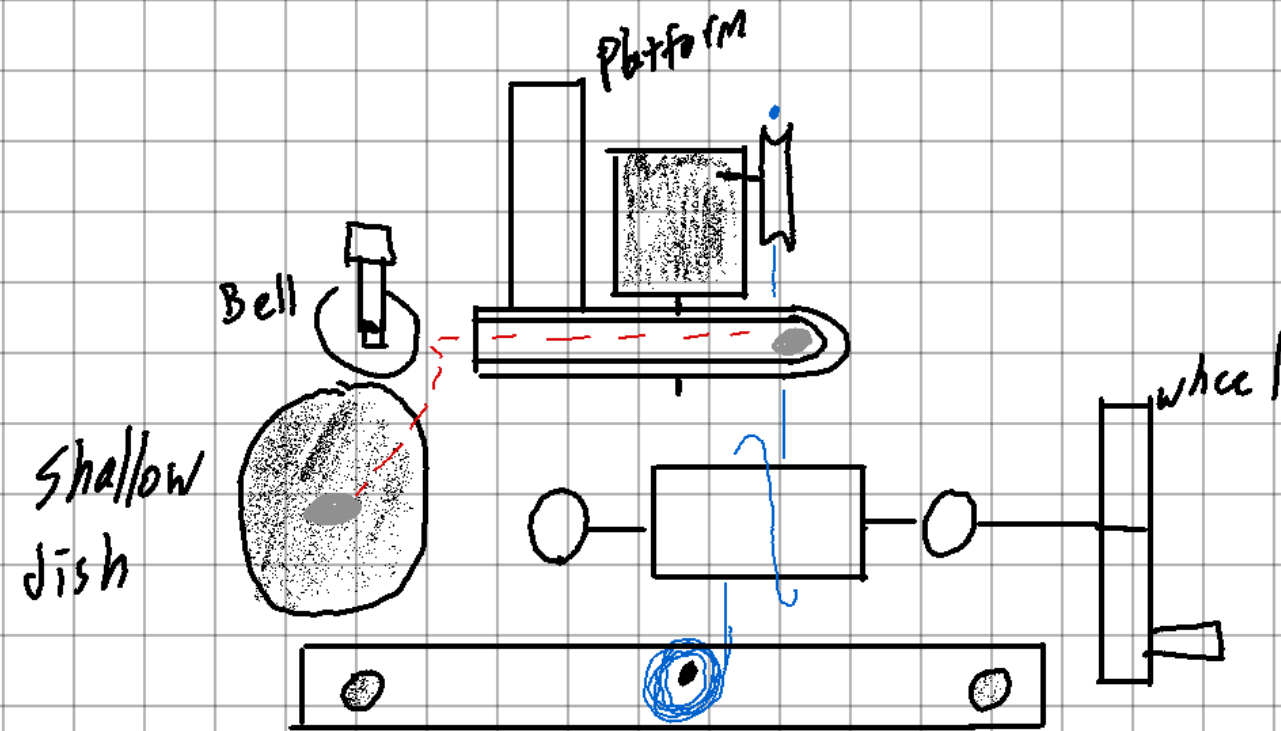
Steeping Complete Notification Contraption

- This contraption is designed to strike a bell one time when steeping is complete, after approximately 5 minutes.
- The weight descends until it starts to rest on the platform. When the weight puts a small amount of force on the platform, a track pivots and a ball bearing (or marble) rolls down the track. The ball bearing strikes the bell and is caught in a shallow dish.
- The shallow dish is separate from the rest of the mechanism so it can be easily repositioned as necessary to catch the ball. It will likely include some padding to help catch the ball. If necessary, a small magnet can be placed under the padding to help retain the ball bearing.
- The inspiration for the bell notification mechanism came from video below. The whole video is worth watching, but skip to 19:30 if you just want to see the bell mechanism.
- <https://www.youtube.com/watch?v=oJzy1vk2zyc&list=PLxYRkuI9JZoOfj6--7C8BmSLWSMhE7KK8&index=25&pp=gAQBiAQB>

Steeping complete notification contraption - overview

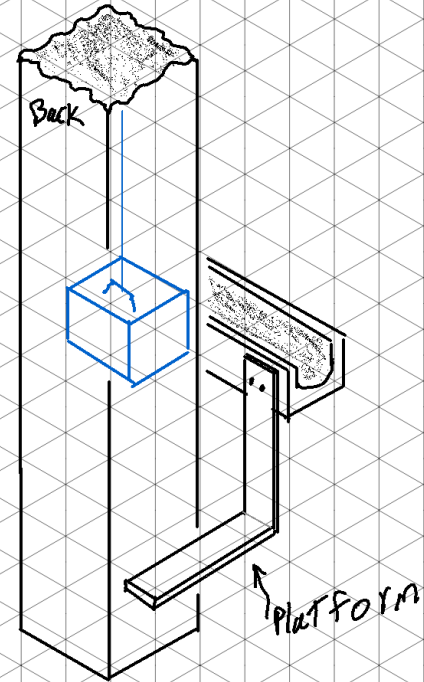


Top Down View



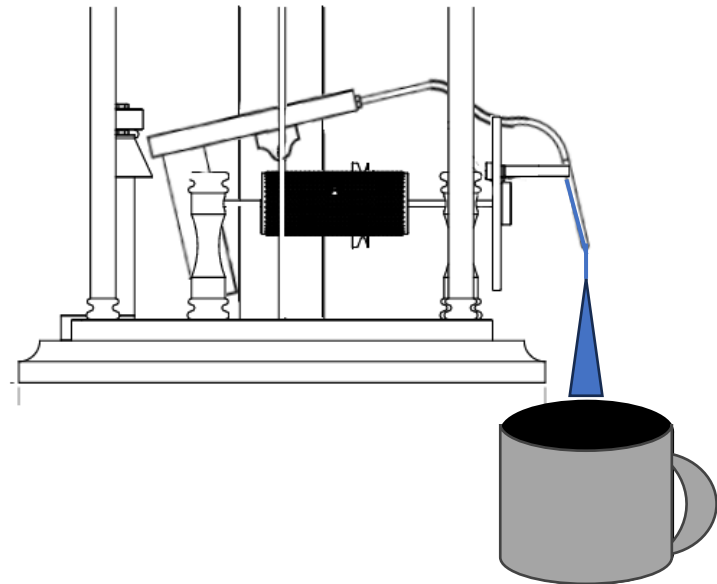
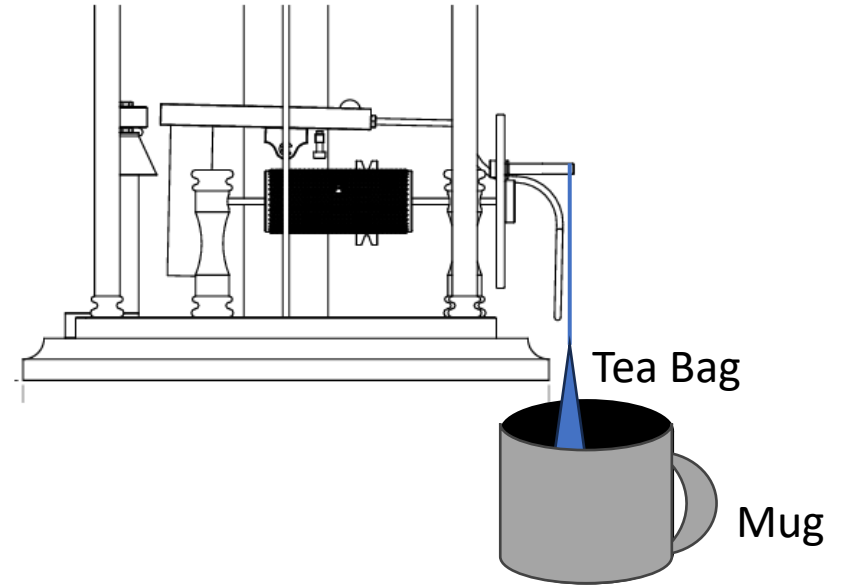
upper arms removed

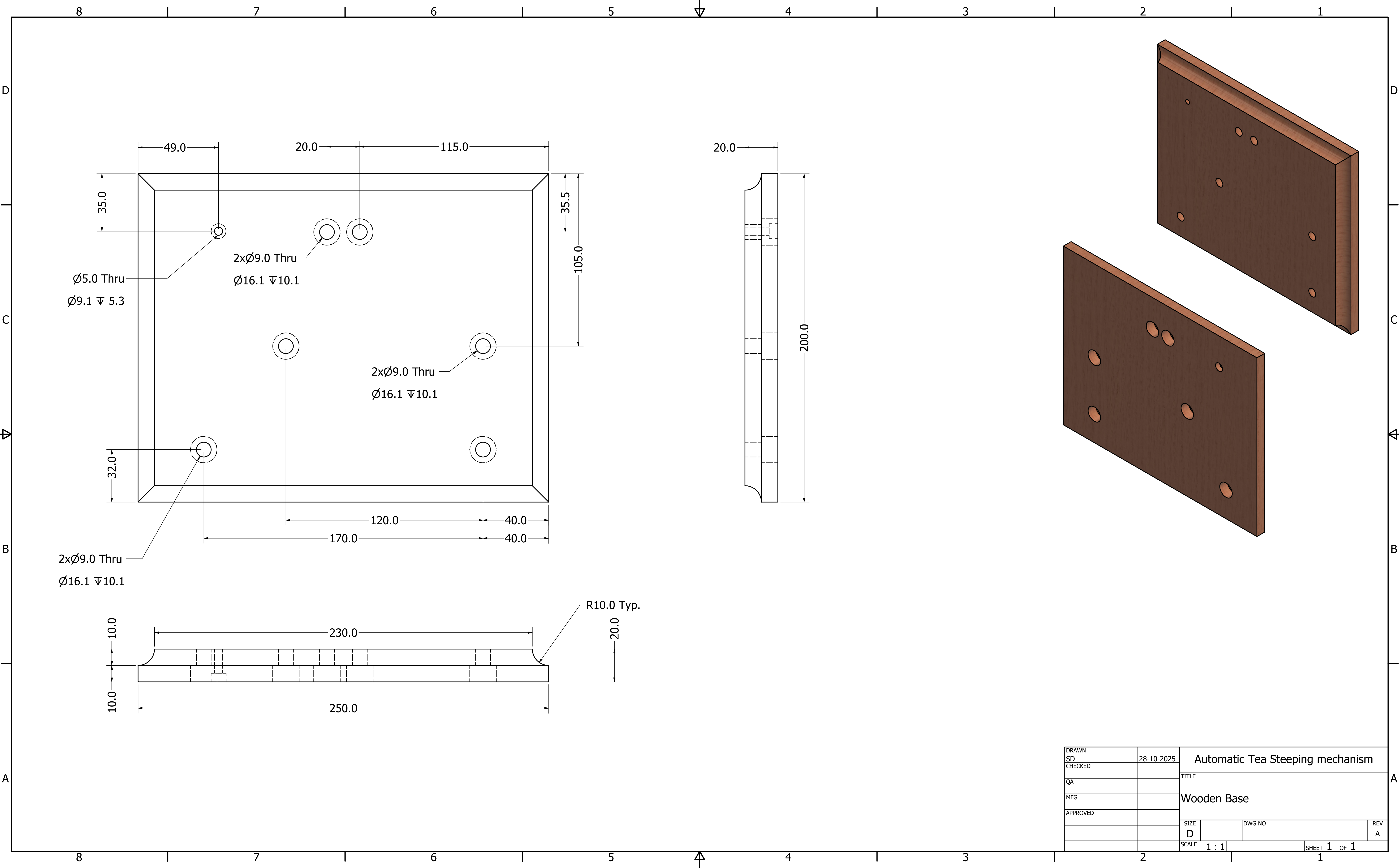
Reverse Perspective of Steeping Complete Notification contraption



Tea Bag Lifter Mechanism

- Attached to the back of the ball track is a bent rod that is the tea bag lifter mechanism
- When the ball track rotates, the rod will rotate up, lifting the tea bag out of the mug





Wooden Base

- Red Morrel Burl - <https://www.youtube.com/watch?v=vFXsJNO3qsl>
- Curly Koa is another good wood for a base
- But Walnut trophy bases are much less expensive:
 - <https://www.jpplus.com/solid-walnut-plaque-base?sku=XP5114-EA>
 - <https://www.jpplus.com/roman-cove-edge-plaque?sku=W791-EA>



Tea Organizer base

In order to have enough room for a mug underneath the wheel, I think the device might need to be placed on a stand like this [tea organizer](#).

Alternatively , this type of chest/organizer could be built into the wooden base. It would be much more elegant this way, though difficult to find an artisan who can make a nice chest and do the machining.



String and Weight

- <https://www.amazon.com/Slotted-Weight-Set-100g-Removable/dp/B08JM71KVX>



Parameters influencing steeping time

- Length of main weight travel
 - Design goal: ~300 mm

$$5 \text{ min} * 60 \frac{\text{sec}}{\text{min}} = \frac{\text{seconds}}{\text{rev}} * \frac{\text{Length of Weight Travel}(\text{mm})}{\text{string extended per rev} \left(\frac{\text{mm}}{\text{rev}} \right)}$$

- Seconds per rev (s/rev)
 - Design goal: $\geq 9 \text{ sec/rev}$
 - Pendulum string length has the largest influence on this parameter
 - Other factors include:
 - Pendulum weight
 - Mass of main weight
 - Distance between static arms (?)
 - Angle between upper and lower static arms
 - Friction
- String extended per rev (mm/rev)
 - Diameter of rotating rod (3mm)
 - Diameter * π

$$300 \text{ sec} \approx 10 \frac{\text{sec}}{\text{rev}} * \frac{300 \text{ mm}}{3\pi \left(\frac{\text{mm}}{\text{rev}} \right)}$$